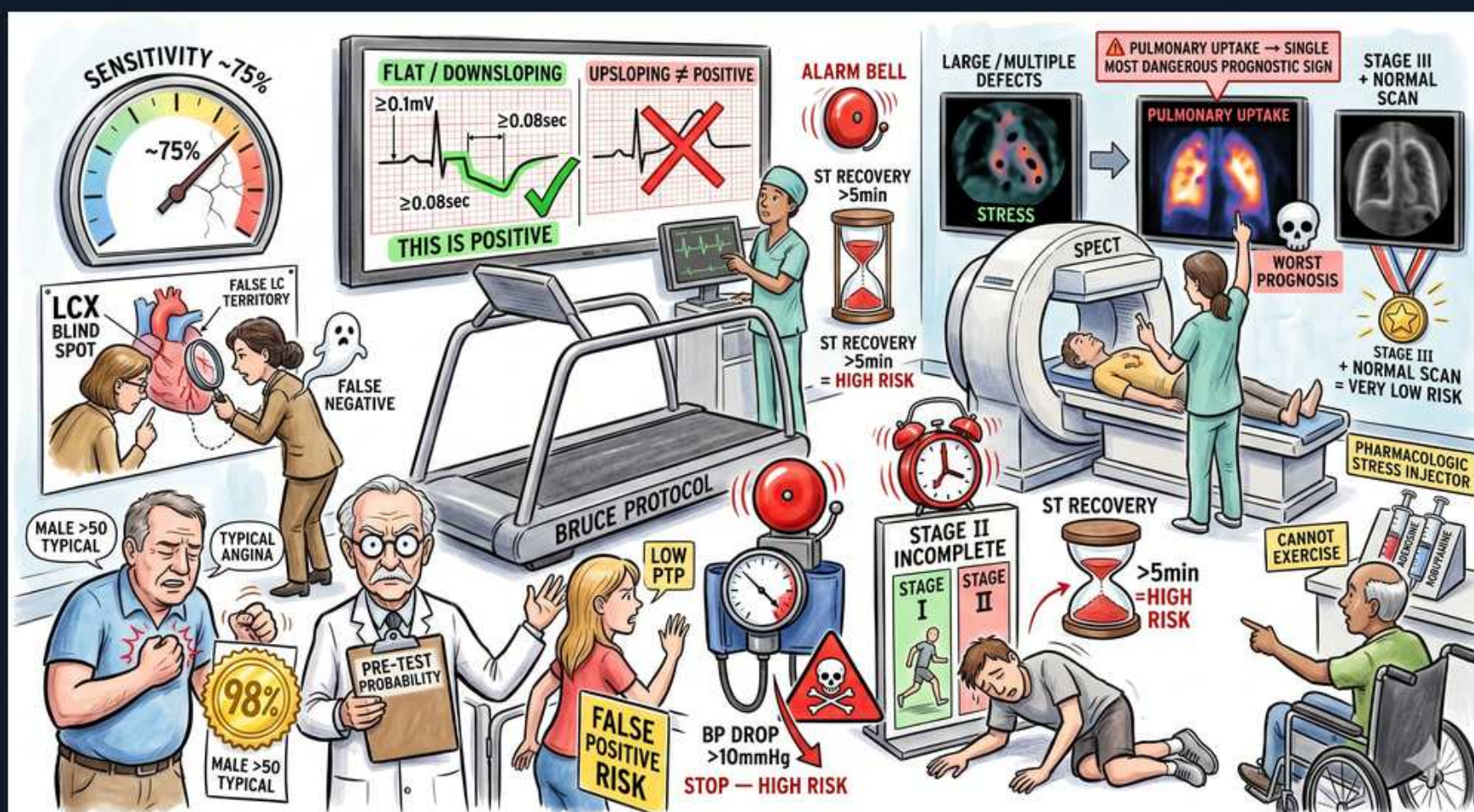


Stress Test #1: Pre-Test Probability & High-Risk Response Markers



1. Sensitivity ~75%

WHAT YOU SEE

Gauge dial ~75% sensitivity

WHAT IT ENCODES

Exercise ECG has ~75% sensitivity, so a normal result cannot exclude CAD — about a quarter of true disease is missed, especially single-vessel and circumflex territory. Add imaging when clinical suspicion remains high.

BOARD TRAP

Ruling out CAD on a negative exercise ECG — ~25% false-negative rate; high suspicion still needs imaging/angiography.

2. Flat/downsloping $\geq 1\text{mm}$ = positive

WHAT YOU SEE

FLAT/DOWNSLOPING $\geq 0.1\text{mV}$ at $\geq 0.08\text{sec}$, 'THIS IS POSITIVE'

WHAT IT ENCODES

A positive exercise ECG is $\geq 1\text{mm}$ (0.1mV) horizontal or downsloping ST depression measured 0.08s past the J-point. Upsloping depression does not count. This is the objective threshold for an ischemic ECG response.

BOARD TRAP

Upsloping ST depression read as positive — only horizontal/downsloping $\geq 1\text{mm}$ at 80ms past J-point is diagnostic.

3. LCx blind spot → false negative

WHAT YOU SEE

LCX BLIND SPOT heart, 'false negative'

WHAT IT ENCODES

The standard 12-lead ECG poorly represents the posterolateral wall, so left circumflex-territory ischemia is a classic source of false-negative exercise ECG. A normal ECG response does not exclude LCx disease.

BOARD TRAP

Trusting a normal exercise ECG to exclude circumflex disease — the posterolateral blind spot causes false negatives.

4. Male >50 with typical angina ~98% PTP

WHAT YOU SEE

Male >50, TYPICAL ANGINA, 98% gold badge

WHAT IT ENCODES

A man over 50 with typical angina has very high pre-test probability (~90%+). With such high prior probability, a negative stress test cannot meaningfully lower post-test probability — proceed toward angiography rather than stress testing.

BOARD TRAP

Stress-testing a high-PTP man with typical angina — a normal test won't reassure; go to coronary angiography.

5. Low PTP → false-positive risk

WHAT YOU SEE

PRE-TEST PROBABILITY, FALSE POSITIVE RISK, LOW PTP

WHAT IT ENCODES

Stress testing is most useful at intermediate pre-test probability. At low PTP most positives are false; at very high PTP most negatives are false. Pre-test probability, not the raw result, determines how to interpret and act.

BOARD TRAP

Applying stress tests across all PTP — they only refine intermediate probability; extremes mislead in both directions.

6. Bruce protocol treadmill

WHAT YOU SEE

BRUCE PROTOCOL treadmill

WHAT IT ENCODES

The Bruce protocol is the standard incremental treadmill test; a diagnostic study requires $\geq 85\%$ of maximum predicted heart rate. Exercise duration and the Duke treadmill score are independently prognostic.

BOARD TRAP

Calling a submaximal (<85% MPPHR) test negative — inadequate workload yields a non-diagnostic, falsely reassuring result.

7. BP drop >10 mmHg → stop, high risk

WHAT YOU SEE

Gauge 'LOW PTP / BP DROP >10 mmHg STOP — HIGH RISK'

WHAT IT ENCODES

A fall in systolic BP >10 mmHg during exercise indicates exertional LV dysfunction from severe multivessel or left main disease — a high-risk sign mandating test termination and prompt angiography.

BOARD TRAP

Pushing the test through exertional hypotension — falling systolic BP signals severe ischemia; stop immediately.

8. Alarm bell — ST recovery >5 min

WHAT YOU SEE

ALARM BELL, 'ST recovery >5 min'

WHAT IT ENCODES

ST depression persisting more than 5 minutes into recovery reflects extensive ischemia and is a high-risk finding that should trigger angiography. Prolonged recovery time tracks with more severe coronary anatomy.

BOARD TRAP

Treating slow ST recovery as benign — depression lasting >5 min into recovery is a high-risk prognostic marker.

9. Stage II incomplete = high risk

WHAT YOU SEE

STAGE I/II ladder, 'STAGE II INCOMPLETE, ST recovery >5 min = HIGH RISK'

WHAT IT ENCODES

Inability to complete even Bruce stage II (low workload) indicates poor functional capacity and is associated with high risk and adverse prognosis. Low exercise tolerance is itself a powerful predictor of cardiac events.

BOARD TRAP

Reassuring a patient who fails early-stage exercise — poor functional capacity (failing stage II) is high risk, not deconditioning alone.

10. Pulmonary uptake → worst prognosis

WHAT YOU SEE

Lung perfusion 'PULMONARY UPTAKE → single most dangerous prognostic sign'

WHAT IT ENCODES

Increased lung uptake of perfusion tracer reflects stress-induced LV dysfunction and elevated filling pressures — the single most dangerous prognostic marker, indicating extensive ischemia and high event risk.

BOARD TRAP

Focusing only on defect size — pulmonary tracer uptake is the strongest high-risk prognostic flag on perfusion imaging.

11. Stage III + normal scan = very low risk

WHAT YOU SEE

STAGE III + NORMAL SCAN, medal 'VERY LOW RISK'

WHAT IT ENCODES

Achieving Bruce stage III with a normal perfusion scan and good exercise capacity confers very low risk and excellent prognosis — no further testing or revascularization required. High functional capacity is strongly protective.

BOARD TRAP

Ordering angiography after a high-workload normal scan — excellent exercise tolerance with normal imaging is reassuringly low risk.

12. Cannot exercise → pharmacologic stress

WHAT YOU SEE

Wheelchair, 'CANNOT EXERCISE → pharmacologic stress injector'

WHAT IT ENCODES

Patients who cannot exercise adequately need pharmacologic stress: vasodilators (adenosine/regadenoson/dipyridamole) with perfusion imaging, or dobutamine for stress echo. Exercise is preferred when possible because capacity adds prognostic data.

BOARD TRAP

Defaulting to exercise testing in patients who cannot walk adequately — submaximal effort is non-diagnostic; use pharmacologic stress.
